



# Artificial Intelligence and Machine Learning

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## AI Landscape

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### Feature Extraction - Principal Component Analysis

It reduces the dimensions into components. **Components are the derivatives of the original features**

It combines the features based on their significance

Each of the component after combining is independent of one another

### Feature Extraction

```
from sklearn.decomposition import PCA  
  
pca = PCA(n_components=2)  
  
x=df.iloc[:,0:4]  
  
principalComponents = pca.fit_transform(x)
```

## Train Test Split

| sepal_length | sepal_width | petal_length | petal_width | Iris_class |
|--------------|-------------|--------------|-------------|------------|
| 5            | 2           | 3.5          | 1           | versicolor |
| 6            | 2.2         | 4            | 1           | versicolor |
| 6.2          | 2.2         | 4.5          | 1.5         | versicolor |
| 6            | 2.2         | 5            | 1.5         | virginica  |
| 4.5          | 2.3         | 1.3          | 0.3         | setosa     |
| 5.5          | 2.3         | 4            | 1.3         | versicolor |
| 6.3          | 2.3         | 4.4          | 1.3         | versicolor |
| 5            | 2.3         | 3.3          | 1           | versicolor |
| 4.9          | 2.4         | 3.3          | 1           | versicolor |
| 5.5          | 2.4         | 3.8          | 1.1         | versicolor |
| 5.5          | 2.4         | 3.7          | 1           | versicolor |
| 5.6          | 2.5         | 3.9          | 1.1         | versicolor |
| 6.3          | 2.5         | 4.9          | 1.5         | versicolor |
| 5.5          | 2.5         | 4            | 1.3         | versicolor |
| 5.1          | 2.5         | 3            | 1.1         | versicolor |
| 4.9          | 2.5         | 4.5          | 1.7         | virginica  |
| 5.7          | 2.5         | 5            | 2           | virginica  |
| 6.3          | 2.5         | 5            | 1.9         | virginica  |
| 5.7          | 2.6         | 3.5          | 1           | versicolor |
| 5.5          | 2.6         | 4.4          | 1.2         | versicolor |
| 5.8          | 2.6         | 4            | 1.2         | versicolor |

Training data

Test data

Train-Test 80:20 or 70:30 or 60:40

## Train Test Split

```
from sklearn.model_selection import train_test_split
```

```
X=df3.iloc[:,0:4] # learning features
```

```
y=df3['Species'] # target feature
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.20,  
random_state=1,shuffle=True)
```

`stratify= class_label` as parameter helps to take the train sample equal in all category of labels

### Iris Dataset – Train Test split

150 – observations --- 120 in train and 30 in test

X\_train – (120,4) X\_test – (30,4)

Y\_train – (120,) y\_test – (30,)

| sepal_length | sepal_width | petal_length | petal_width | Iris_class |
|--------------|-------------|--------------|-------------|------------|
| 5.1          | 3.5         | 1.4          | 0.1         | versicolor |
| 4.9          | 3.6         | 1.4          | 0.1         | versicolor |
| 6.2          | 2.2         | 4.5          | 1.5         | versicolor |
| 6            | 2.2         | 5            | 1.5         | virginica  |
| 4.5          | 2.3         | 1.3          | 0.3         | setosa     |
| 5.5          | 2.3         | 4            | 1.3         | versicolor |
| 6.3          | 2.3         | 4.4          | 1.3         | versicolor |
| 5            | 2.3         | 3.3          | 1           | versicolor |
| 4.9          | 2.4         | 3.3          | 1           | versicolor |
| 5.5          | 2.4         | 3.8          | 1.1         | versicolor |
| 5.5          | 2.4         | 3.7          | 1           | versicolor |
| 5.6          | 2.5         | 3.9          | 1.1         | versicolor |
| 6.3          | 2.5         | 4.9          | 1.5         | versicolor |
| 5.5          | 2.5         | 4            | 1.3         | versicolor |
| 5.1          | 2.5         | 3            | 1.1         | versicolor |
| 4.9          | 2.5         | 4.5          | 1.7         | virginica  |
| 5.7          | 2.5         | 5            | 1.9         | virginica  |
| 6.3          | 2.5         | 5            | 1.9         | virginica  |
| 5.7          | 2.6         | 3.5          | 1           | versicolor |
| 5.5          | 2.6         | 4.4          | 1.2         | versicolor |
| 5.8          | 2.6         | 4            | 1.2         | versicolor |

Training data

X\_Train

Y\_Train

Y\_test

Test data

X\_test



# THANK YOU

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